

# DEMAND AND SUPPLY

Lecture 3



- ▶ **Market-** An arrangement that brings buyers and sellers together to do business with each other, i.e. to exchange goods and services.
- ▶ Different types of Markets:
  - Market for goods
  - Market for services
  - Factor market
  - Money market...etc. etc.

# MARKETS

- ▶ **Competitive Market-** a market that has many buyers and many sellers, so **NO single buyer or seller can influence the price.**
- ▶ Example- Market for biscuits, vegetables etc.

## A COMPETITIVE MARKET

- ▶ Buyers determine *demand*.
- ▶ Sellers determine *supply*



# MARKETS AND COMPETITION

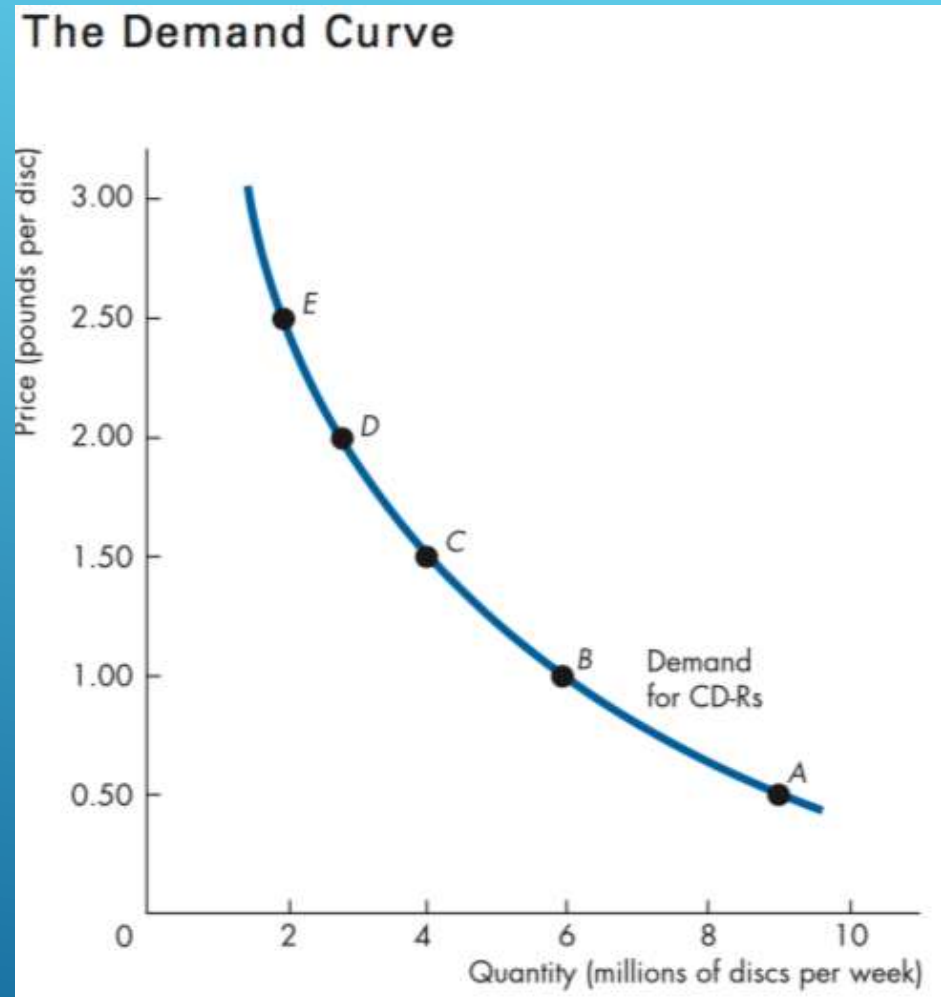
- ▶ **Demand** in economics is defined as consumers' willingness and ability to consume a given good.
- ▶ *Quantity demanded* is the amount of a good that buyers are willing and able to purchase.
- ▶ **The Law of Demand-** The higher the price of a good, the smaller is the quantity demanded; and the lower the price of a good the greater is the quantity demanded, other things remaining same (ceteris paribus).

DEMAND & SUPPLY- DEMAND

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# DEMAND CURVE

- ▶ **Demand Curve-** The inverse relationship between price and quantity demanded of a good (the law of demand) is represented by the demand curve.
- ▶ **Quantity demanded-** the quantity demanded of a good or service is the amount that consumers plan to buy during a given time period at a particular price. It refers to a point on a demand curve.



- ▶ **Change in quantity demanded-** Movement along the demand curve: when price of a good changes and everything remains the same.
- ▶ **Change in demand-** When there is a shift in demand curve : when the price remains same but some other influence on buyers' plans changes.

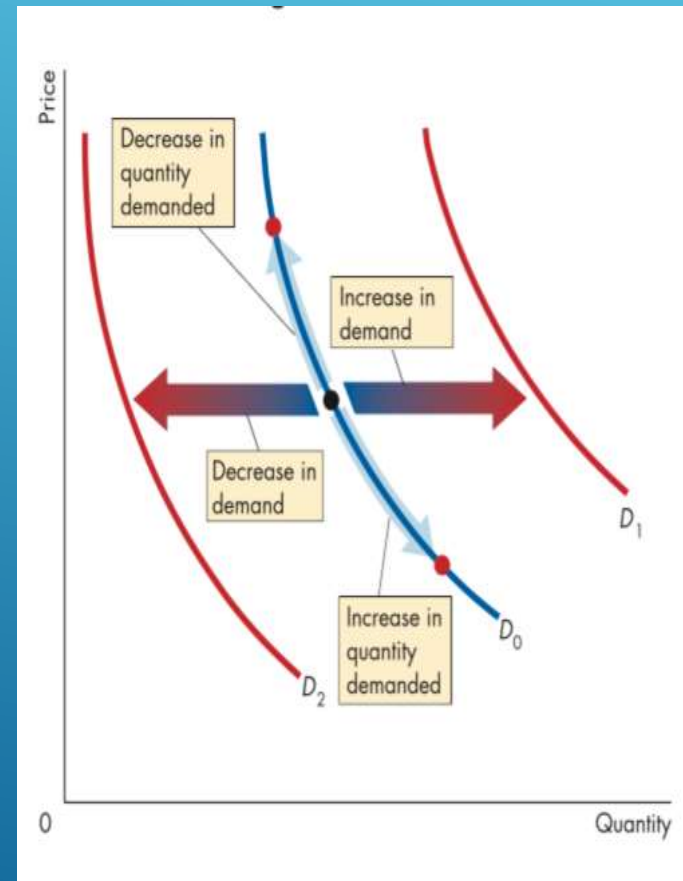
## CHANGE IN QUANTITY DEMANDED VERSUS CHANGE IN DEMAND

When any factor that influences buying plans other than the price of the good changes, there is a *change in demand*.

► **Factors that change in demand:**

1. Prices of related goods (Substitutes and Complements)
2. Expected future prices
3. Income (normal good and inferior good).
4. Expected future income (or easier credit)
5. Population (size and age structure)
6. Preferences

## CHANGE IN DEMAND





If a firm supplies a good or a service, the firm

1. Has the resources and technology to produce it,
2. Can profit from producing it, and
3. Plans to produce it and sell it.

► Resources and technology determine what is possible to produce.

# SUPPLY

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- ▶ The quantity supplied of a good or service is the amount that producers plan to sell during a given time period at a particular price.
- ▶ Supply refers to the entire relationship between the quantity supplied and the price of a good

QUANTITY SUPPLIED

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- ▶ The higher the price of a good, the greater the quantity supplied. The lower the price of a good the smaller is the quantity supplied; other things remaining the same.

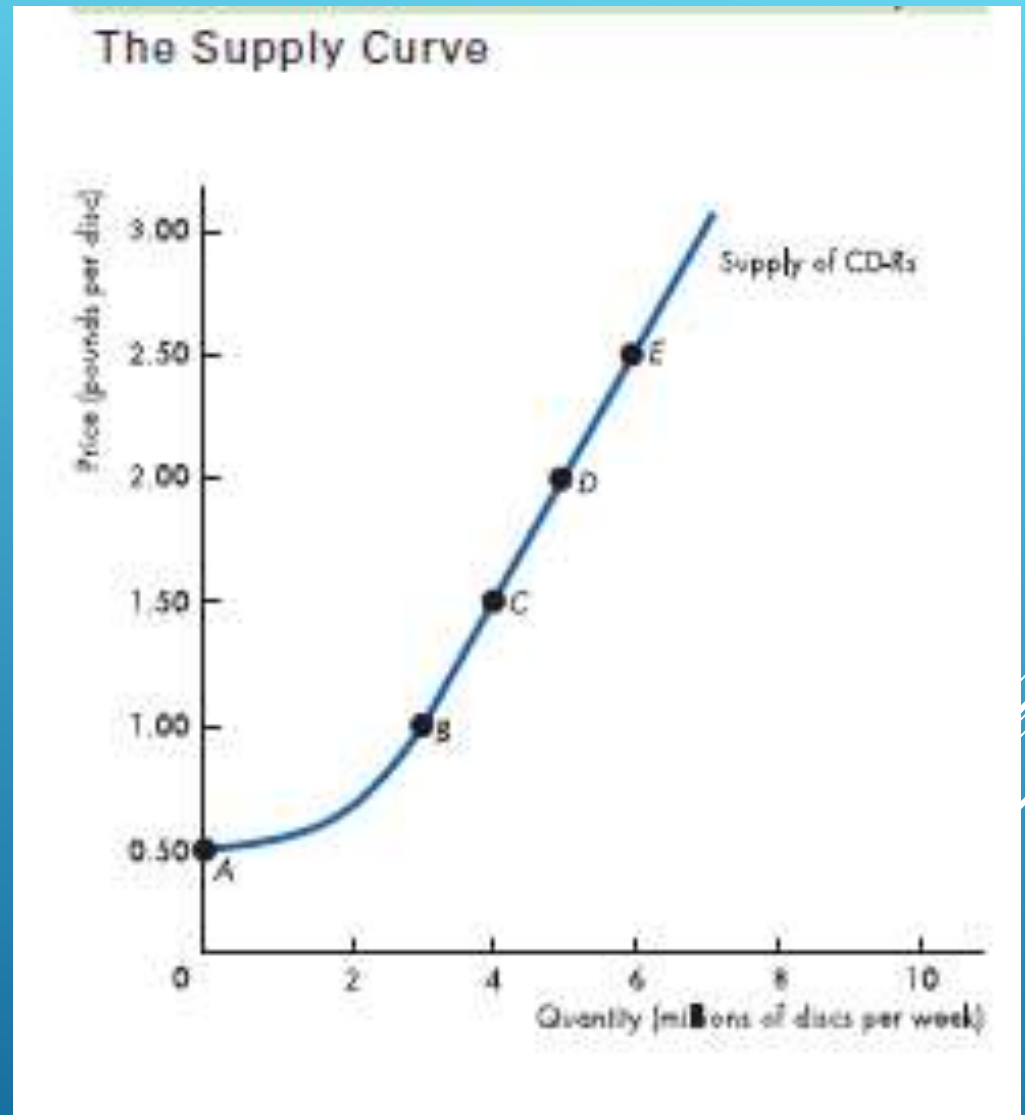
## Why?

- ▶ As the better the price of a good, a business/firm is willing to produce more of it
- ▶ As the quantity produced of any good increases, the marginal cost of producing the good increases

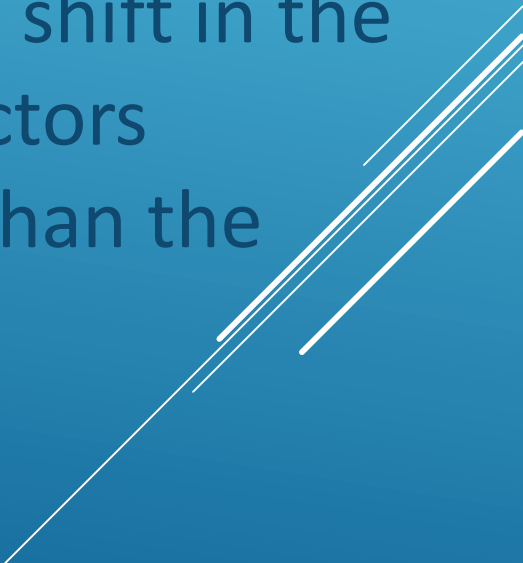
# THE LAW OF SUPPLY

- ▶ The supply curve shows the relationship between the quantity supplied of a good and its price when all other influences on producers' planned sales remain the same.

## SUPPLY CURVE



# CHANGE IN QUANTITY SUPPLIED VS CHANGE IN SUPPLY

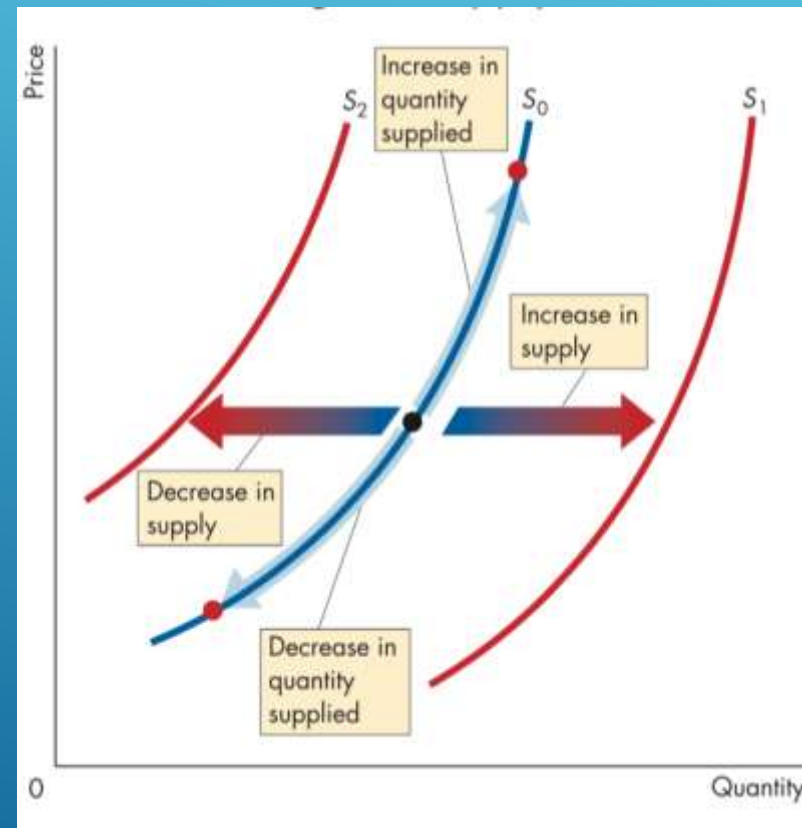
- ▶ **Change in quantity supplied** refers to a point on the supply curve — a particular quantity supplied at a particular price
  - ▶ **Change in supply** is when there is a shift in the supply curve — when any of the factors influencing the selling plans other than the price of the good changes.
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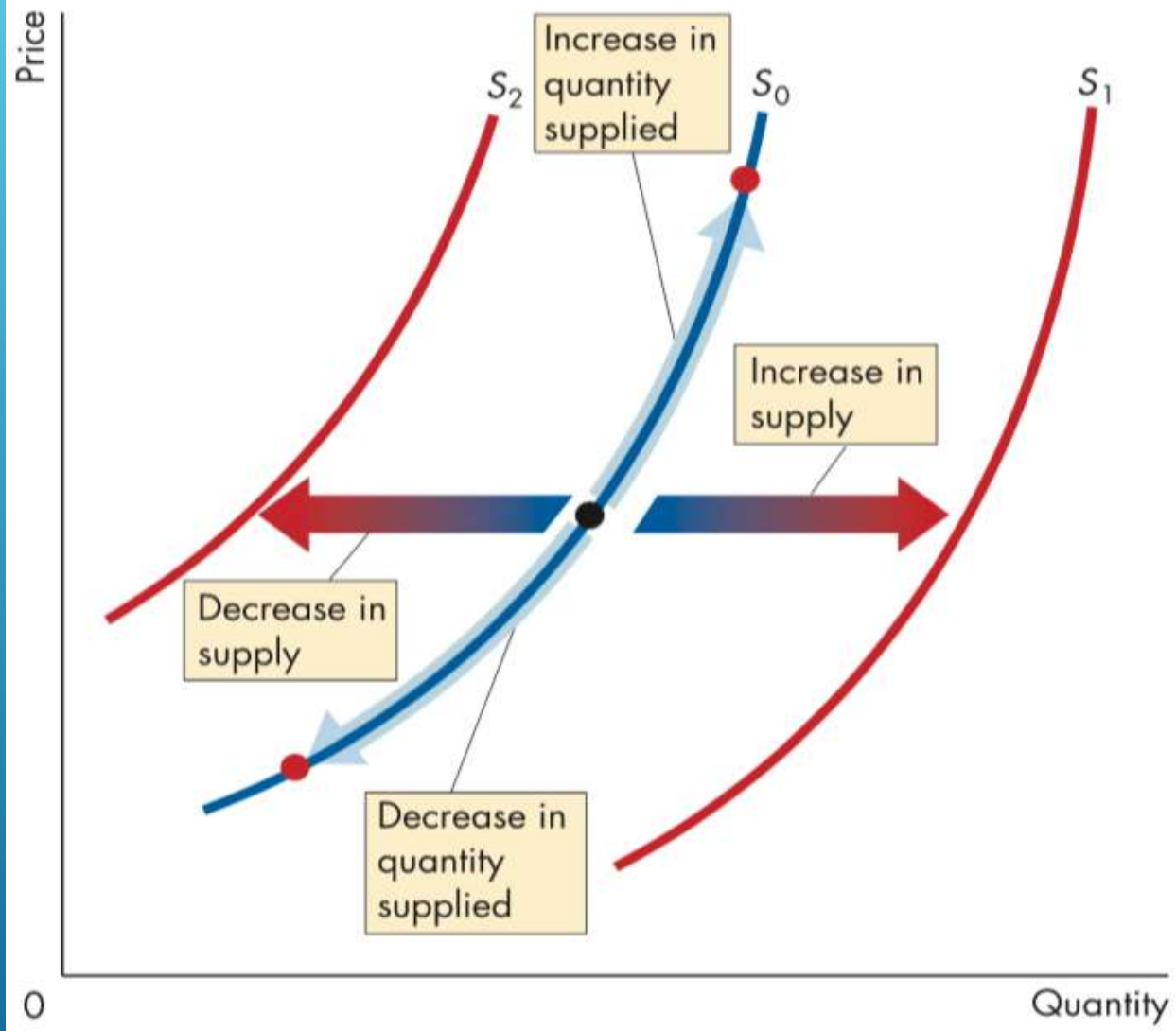
► It is when any of the factors influences the selling plans, other than changes in the price of the good, then we say there is a change of supply.

► Factors that bring changes in supply are:

1. Prices of Factors of Production
2. Prices of related produced goods
3. Expected future prices
4. Number of suppliers
5. Technology
6. The state of nature

## CHANGE IN SUPPLY





# MARKET EQUILIBRIUM

Chapter 3



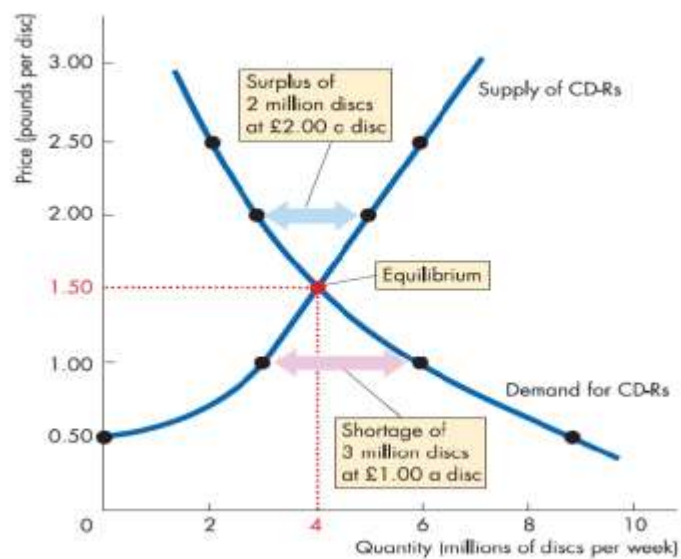
- ▶ The **equilibrium price** is the price at which the quantity demanded equals the quantity supplied.
- ▶ Equilibrium in a market occurs when the price balances the plans of buyers and sellers.
- ▶ The **equilibrium quantity** is the quantity bought and sold at the equilibrium price.
- ▶ A market moves towards its equilibrium because:
  - ▶ Price regulates buying and selling plans.
  - ▶ Price adjusts when plans don't match

# MARKET EQUILIBRIUM

- ▶ The price of a good regulates the quantities demanded and supplied. If the price is too high, the quantity supplied exceeds the quantity demanded.
- ▶ If the price is too low, the quantity demanded exceeds the quantity supplied.
- ▶ There is one price at which the quantity demanded equals the quantity supplied.

## PRICE AS A REGULATOR

## Equilibrium



| Price<br>(pounds<br>per disc) | Quantity<br>demanded | Quantity<br>supplied | Shortage (-)<br>or surplus (+) |
|-------------------------------|----------------------|----------------------|--------------------------------|
| (millions of discs per week)  |                      |                      |                                |
| 0.50                          | 9                    | 0                    | -9                             |
| 1.00                          | 6                    | 3                    | -3                             |
| <b>1.50</b>                   | <b>4</b>             | <b>4</b>             | <b>0</b>                       |
| 2.00                          | 3                    | 5                    | +2                             |
| 2.50                          | 2                    | 6                    | +4                             |

# MARKET EQUILIBRIUM

## ▶ **A Shortage Forces the Price Up:**

- ▶ When the price is £1. Consumers plan to buy 6 million discs a week, and producers plan to sell 3 million discs a week.
- ▶ Producers raise the price in response.
- ▶ As producers push the price up, the price rises towards its equilibrium.
- ▶ The rising price reduces the shortage because it decreases the quantity demanded and increases the quantity supplied.
- ▶ When the price has increased to the point at which there is no longer a shortage, the forces moving the price stop operating and the price comes to rest at its equilibrium.

# PRICE ADJUSTMENTS

## ▶ **A Surplus Forces the Price Down**

- ▶ Suppose the price of a CD-R is £2. Producers plan to sell 5 million discs a week, and consumers plan to buy 3 million discs a week.
- ▶ Some producers, unable to sell the quantities of discs they planned to sell, cut their prices. In addition, some producers reduce production.
- ▶ As producers cut the price, the price falls towards its equilibrium.
- ▶ When the price has fallen to the point at which there is no longer a surplus, the forces moving the price stop operating and the price comes to rest at its equilibrium.

# PRICE ADJUSTMENTS

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- ▶ When the price has fallen to the point at which there is no longer a surplus, the forces moving the price stop operating and the price comes to rest at its equilibrium.

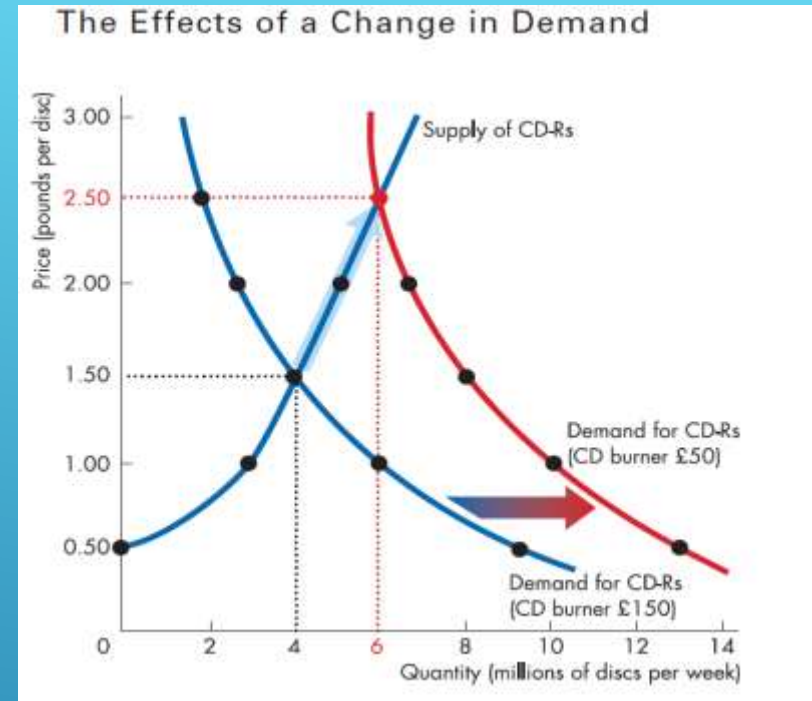
# PRICE ADJUSTMENTS

- ▶ The market equilibrium price is the best deal available for buyers and sellers.
- ▶ At the price at which the quantity demanded equals the quantity supplied neither buyers nor sellers can do business at a better price.
- ▶ The price coordinates the plans of buyers and sellers, and no one has an incentive to change it.

**THE BEST DEAL AVAILABLE FOR BUYERS  
AND SELLERS**

# A CHANGE IN DEMAND

- ▶ Supply curve remains unchanged.
- ▶ Suppose the demand for CDs increase.
- ▶ Demand curve shifts to the right.



# PREDICTING CHANGES IN PRICE AND QUANTITY

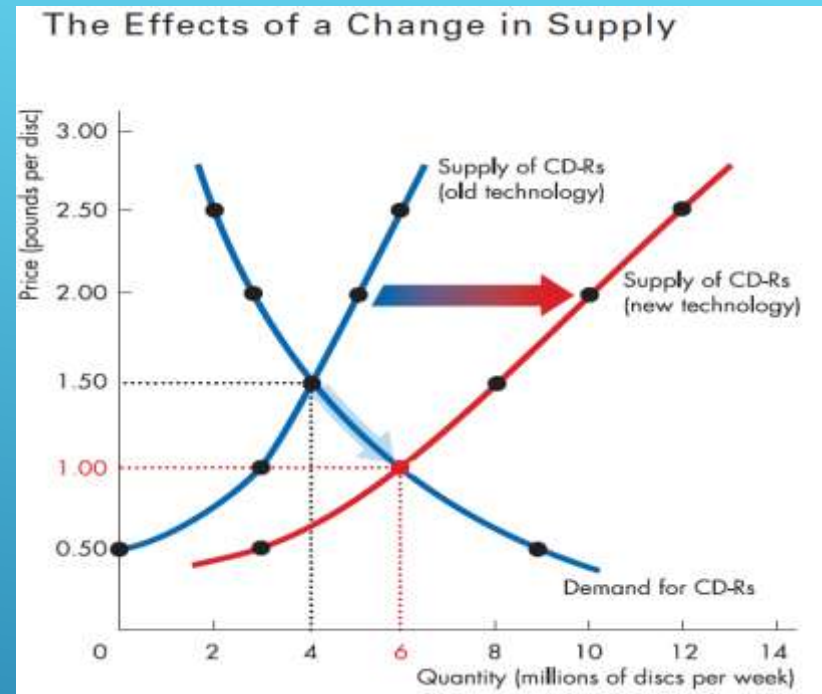


# PREDICTING CHANGES IN PRICE AND QUANTITY

- ▶ We can now make our first two predictions:
    - ▶ 1. When demand increases, both the price and the quantity increase.
    - ▶ 2. When demand decreases, both the price and the quantity decrease.
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## CHANGE IN SUPPLY

- ▶ Suppose the supply for CDs increase.
- ▶ Supply curve shifts to the right.
- ▶ Demand curve remains unchanged



## PREDICTING CHANGES IN PRICE AND QUANTITY

# PREDICTING CHANGES IN PRICE AND QUANTITY

- ▶ We can now make two more predictions:
    - ▶ **1. When supply increases, the quantity increases and the price falls.**
    - ▶ **2. When supply decreases, the quantity decreases and the price rises.**
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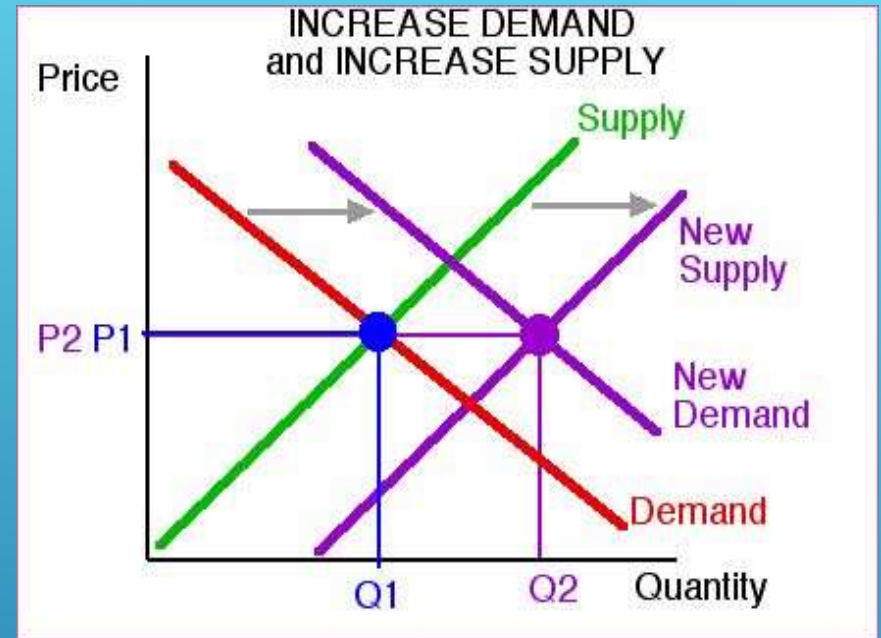
# MARKET EQUILIBRIUM: PREDICTING CHANGES IN PRICE AND QUANTITY

## Chapter 3

- ▶ We've seen that an increase in the demand raises its price and increases the quantity bought and sold.
- ▶ We've seen that an increase in the supply lowers its price and increases the quantity bought and sold.
- ▶ So if **both Demand and Supply increase together**, the quantity increases.
- ▶ However, the effect on price is uncertain.
- ▶ The effects on price depend on the magnitudes of the change in demand and supply

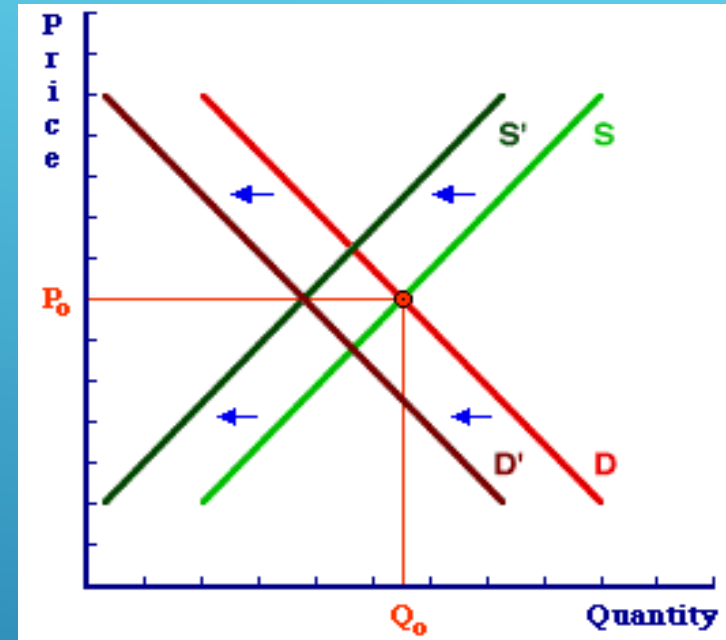
## INCREASE IN BOTH DEMAND AND SUPPLY

- If supply and demand both increase by the same magnitude, then the price does not change.



# INCREASE IN BOTH DEMAND AND SUPPLY

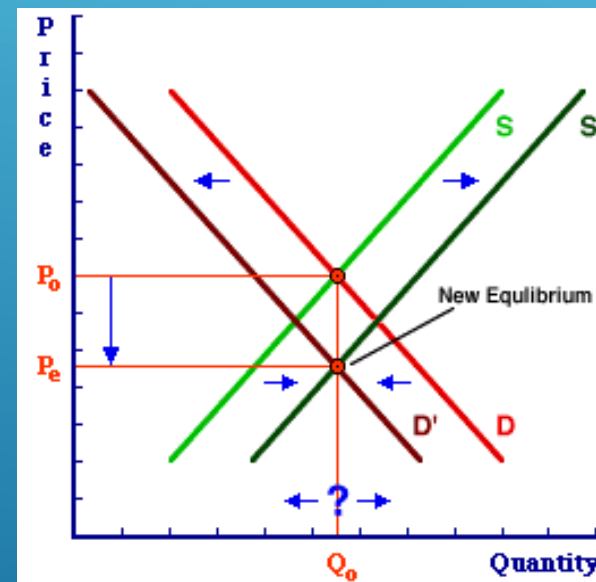
- ▶ If supply and demand both decrease by the same magnitude, then the price does not change.



# DECREASE IN BOTH DEMAND AND SUPPLY

# DECREASE IN DEMAND AND INCREASE IN SUPPLY

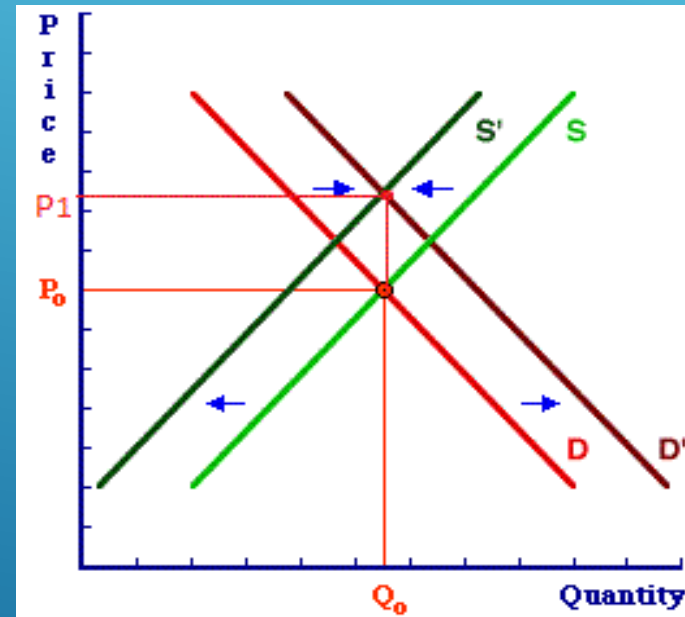
- ▶ If demand decreases and supply increases, there is a fall in price.
- ▶ If the changes in demand and supply is by the same magnitude, then there is no change in quantity.
- ▶ If the magnitude of change in demand or supply varies even slightly, there will be a change in the quantity.





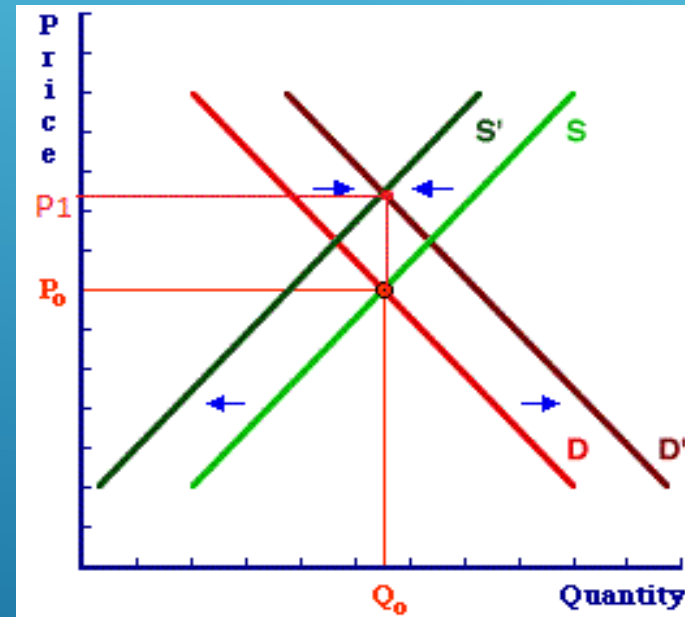
# INCREASE IN DEMAND AND DECREASE IN SUPPLY

- ▶ If demand increases and supply decreases, there is a rise in price.
- ▶ If the changes in demand and supply is by the same magnitude, then there is no change in quantity.
- ▶ If the magnitude of change in demand or supply varies even slightly, there will be a change in the quantity.



# INCREASE IN DEMAND AND DECREASE IN SUPPLY

- ▶ If demand increases and supply decreases, there is a rise in price.
- ▶ If the changes in demand and supply is by the same magnitude, then there is no change in quantity.
- ▶ If the magnitude of change in demand or supply varies even slightly, there will be a change in the quantity.



Required Readings for Today!  
*Parkin, Powell, Matthews.*  
*Economics. 8<sup>th</sup> Edition*  
*Chapter 3- Demand & Supply*

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